

Feedback — Module 1 Quiz

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You submitted this quiz on **Sun 13 Sep 2015 10:24 AM PDT**. You got a score of **8.00** out of **10.00**.

Question 1

Reproducibility is defined informally as the ability to recompute data analytic results conditional on an observed data set and knowledge of the statistical pipeline used to calculate them [Peng 2011, Science](#). Replicability of a study is the chance that a new experiment targeting the same scientific question will produce a consistent result [Asendorpf 2013 European Journal of Personality](#).

Susan asks Joe for his data shared according to the data sharing plan discussed in the lectures. Which of the following are reasons the study may be reproducible, but not replicable?

Your Answer	Score	Explanation
☐ Susan only uses Python and Joe uses R so when she runs a new experiment, she will definitely get different results.		
☒ The identified effect can be reproduced from Joe's code and data, but may be due only to random variation and not appear in future studies.	✓ 1.00	
☐ The code for processing the data is written as a collection of R scripts and not R markdown files, so it is difficult to understand what each step does.		
☐ Joe doesn't make the raw data accessible so Susan can't re-run his code.		
Total	1.00 / 1.00	

Question 2

Put the following code chunk at the top of an R markdown document called *test.Rmd* but set

```
eval=TRUE
```

```
```{r setup, eval=FALSE}
knitr::opts_chunk$set(cache=TRUE)
```
```

Then create the following code chunks

```
```{r }
x = rnorm(10)
plot(x,pch=19,col="dodgerblue")
```
```

```
```{r }
y = rbinom(20,size=1,prob=0.5)
table(y)
```
```

| Your Answer | Score | Explanation |
|--|-------------|-------------|
| ☐ The table is random each time you knit the document, but the plot is always the same after you knit it the first time. | | |
| ☒ The plot is random the first time you knit the document. It is identical to the first time the second time you knit the document. After removing the folders <code>test_cache</code> and <code>test_files</code> they generate new random versions. | ✓ 1.00 | |
| ☐ The plot and table are random the first time you knit the document. They are identical the second time you knit the document. After removing the folders <code>test_cache</code> and <code>test_files</code> they are still identical. | | |
| ☐ The plot is random each time you knit the document, but the table is always the same after you knit it the first time. | | |
| Total | 1.00 / 1.00 | |

Question 3

Create a `summarizedExperiment` object with the following code

```
library(Biobase)
library(GenomicRanges)
data(sample.ExpressionSet, package = "Biobase")
```

```
se = makeSummarizedExperimentFromExpressionSet(sample.ExpressionSet)
```

Look up the help files for `summarizedExperiment` with the code `?summarizedExperiment`. How do you access the genomic data for this object? How do you access the phenotype table? How do you access the feature data? What is the unique additional information provided by `rowRanges(se)`?

| Your Answer | Score | Explanation |
|--|-----------------------------|-------------|
| <p>☒ Get the genomic table with <code>assay(se)</code>, get the phenotype table with <code>colData(se)</code>, get the feature data with <code>rowData(se)</code>. <code>rowRanges(se)</code> gives information on the genomic location and structure of the measured features.</p> | <div><div>✓</div>1.00</div> | |
| <p>☐ Get the genomic table with <code>assay(se)</code>, get the phenotype table with <code>colData(se)</code>, get the feature data with <code>rowData(se)</code>. <code>rowRanges(se)</code> gives the range of possible values for the expression data.</p> | | |
| <p>☐ Get the genomic table with <code>exprs(se)</code>, get the phenotype table with <code>colData(se)</code>, get the feature data with <code>rowData(se)</code>. <code>rowRanges(se)</code> gives information on the genomic location and structure of the measured features</p> | | |
| <p>☐ Get the genomic table with <code>exprs(se)</code>, get the phenotype table with <code>colData(se)</code>, get the feature data with <code>rowData(se)</code>. <code>rowRanges(se)</code> gives the range of possible values for the expression data.</p> | | |
| Total | 1.00 / 1.00 | |

Question 4

Suppose that you have measured ChIP-Seq data from 10 healthy individuals and 10 metastatic cancer patients. For each individual you split the sample into two identical sub-samples and perform the ChIP-Seq experiment on each sub-sample. How can you measure (a) biological variability, (b) technical variability and (c) phenotype variability.

| Your Answer | Score | Explanation |
|---|-------|-------------|
| <p>☐ (a) By looking at variation across samples from 10 different</p> | | |

individuals with cancer (b) By looking at variation across samples from 10 different healthy individuals (c) By comparing the average measurements on the healthy individuals to the measurements on the individuals with cancer.

☐ (a) By looking at variation across samples from 10 different individuals with cancer (b) By looking at variability between the measurements on the two sub-samples from the same sample and (c) by comparing the average measurements on the healthy individuals to the measurements on the individuals with cancer.

☐ (a) By looking at variation across samples from 10 different individuals with cancer (b) By comparing the average variability in the cancer and normal individuals (c) By comparing the average measurements on the healthy individuals to the measurements on the individuals with cancer.

☒ (a) & (c) By looking at variation across samples from 10 different healthy individuals. (b) by comparing the average measurements on the healthy individuals to the measurements on the individuals with cancer. ✖ 0.00

| | |
|-------|--------|
| Total | 0.00 / |
| | 1.00 |

Question 5

Load the Bottomly and the Bodymap data sets with the following code:

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata_bot=pData(bot)

con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata_bm=pData(bm)
```

Just considering the phenotype data what are some reasons that the Bottomly data set is likely a better experimental design than the Bodymap data? Imagine the question of interest in the Bottomly data is to compare strains and in the Bodymap data it is to compare tissues.

Your Answer**Score****Explanation**

- ☐ The Bodymap data has more technical replicates than the Bottomly data.
- ☐ The Bodymap data has measured more levels of the outcome of interest (tissues) than the Bottomly data has measured (strains).
- ☒ The covariates in the Bottomly data set (experiment number, lane number) are balanced with respect to strain. The covariates in the Bodymap data set (gender, age, number of technical replicates) are not balanced with respect to tissue.
- ☐ The Bottomly data has a smaller sample size than the Bodymap data.

✓ 1.00

Total

1.00 /

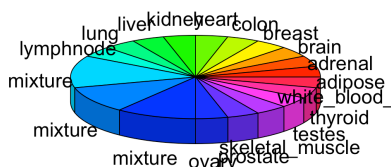
1.00

Question 6

What are some reasons why this plot is not useful for comparing the number of technical replicates by tissue (you may need to install the plotrix package).

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata_bm=pData(bm)

library(plotrix)
pie3D(pdata_bm$num.tech.reps,labels=pdata_bm$tissue.type)
```



Your Answer**Score****Explanation**

☐ The pie chart makes it easier to compare the numbers of replicates, but the 3-d effect is too interesting and draws attention away from the difference.

☒ The plot obscures the fact that most of the samples have 2 technical replicates.

✓ 0.00

☐ The data could much more easily be shown as a heatmap.

☐ There is nothing wrong with the plot, it accurately shows how many replicates of each type there are.

Total

0.00 /

0.00

Question 7

Load the Bottomly data:

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
edata = exprs(bm)
```

Which of the following code chunks will make a heatmap of the 500 most highly expressed genes (as defined by total count), without re-ordering due to clustering? Are the highly expressed samples next to each other in sample order?

Your Answer**Score****Explanation**

```
row_sums = rowSums(edata)
index = which(rank(-row_sums) < 500 )
heatmap(edata[index,],Rowv=NA)
```

The highly expressed samples are not next to each other.



```
row_sums = rowSums(edata)
index = which(rank(-row_sums) < 500 )
heatmap(edata[index,],Rowv=NA)
```

The highly expressed samples are next to each other.



1.00

```
row_sums = rowSums(edata)
edata = edata[order(-row_sums),]
index = 1:500
heatmap(edata[index,],Rowv=NA,Colv=NA)
```

Yes they are.



```
row_sums = rowSums(edata)
edata = edata[order(row_sums),]
index = which(rank(-row_sums) < 500 )
heatmap(edata[index,],Rowv=NA,Colv=NA)
```

No they are not next to each other.

Total

1.00 / 1.00

Question 8

Load the Bodymap data using the following code:

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata = pData(bm)
edata = exprs(bm)
```

Make an MA-plot of the first sample versus the second sample using the log2 transform (hint: you may have to add 1 first) and the `rlog` transform from the DESeq2 package. How are the two MA-plots different? Which kind of genes appear most different in each plot?

| Your Answer | Score | Explanation |
|--|-------|-------------|
| ☐ <p>The plots are very different as the <code>log2</code> plot seems to shrink low abundance genes more and the <code>rlog</code> plot seems to shrink high abundance genes more. The genes in the middle of the distribution show the biggest differences.</p> | | |
| ☐ <p>The plots look pretty similar, but there are two strong diagonal stripes (corresponding to the zero count genes) in the <code>rlog</code> plot. In both cases, the genes in the middle of the expression</p> | | |

Score

Explanation



The plots are very different as the `log2` plot seems to shrink low abundance genes more and the `rlog` plot seems to shrink high abundance genes more. The genes in the middle of the distribution show the biggest differences.



The plots look pretty similar, but there are two strong diagonal stripes (corresponding to the zero count genes) in the `rlog` plot. In both cases, the genes in the middle of the expression

distribution show the biggest differences, but the low abundance genes seem to show smaller differences with the `log2` transform.



✓ 1.00

The plots look pretty similar, but the `rlog` transform seems to shrink the low abundance genes more. In both cases, the genes in the middle of the expression distribution show the biggest differences.



The plots are very different, there are two strong diagonal stripes (corresponding to the zero count genes) in the `log2` plot and the high abundance genes are most different, but the low abundance genes seem to show smaller differences with the `rlog` transform

Total

1.00 /
1.00

Question 9

Load the Montgomery and Pickrell eSet:

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData")
load(file=con)
close(con)
mp = montpick.eset
pdata=pData(mp)
edata=as.data.frame(exprs(mp))
fdata = fData(mp)
```

Cluster the data in three ways:

1. With no changes to the data
2. After filtering all genes with `rowMeans` less than 100
3. After taking the `log2` transform of the data without filtering

Color the samples by which study they came from (Hint: consider using the function

`myplclust.R` in the package `rafalib` available from CRAN and looking at the argument `lab.col`.)

How do the methods compare in terms of how well they cluster the data by study? Why do you think that is?

Your Answer

Score

Explanation

☐ Clustering with or without log2 transform is about the same. Clustering after filtering shows better clustering with respect to the study variable. The reason is that the lowly expressed genes have some extreme outliers that skew the calculation.

☒ Clustering with or without filtering is about the same. Clustering after the log2 transform shows better clustering with respect to the study variable. The likely reason is that the highly skewed distribution doesn't match the Euclidean distance metric being used in the clustering example.

✓ 1.00

☐ Clustering is identical with all three approaches and they show equal clustering. Clustering with or without log2 transform is about the same. Clustering after filtering shows better clustering with respect to the study variable. The reason is that it is just the lowly expressed genes that make the distance hard to calculate.

☐ Clustering is identical with all three approaches and they show equal clustering. The `log2` transform is a monotone transformation so it doesn't affect the clustering.

Total

1.00 /
1.00

Question 10

Load the Montgomery and Pickrell eSet:

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData")
load(file=con)
close(con)
mp = montpick.eset
pdata=pData(mp)
edata=as.data.frame(exprs(mp))
fdata = fData(mp)
```

Cluster the samples using k-means clustering after applying the `log2` transform (be sure to add 1). Set a seed for reproducible results (use `set.seed(1235)`). If you choose two clusters, do you get the same two clusters as you get if you use the `cutree` function to cluster the samples into two groups? Which cluster matches most closely to the study labels?

| Your Answer | Score | Explanation |
|---|-------------|-------------|
| ☒ They produce different answers. The k-means clustering matches study better. Hierarchical clustering would look better if we went farther down the tree but the top split doesn't perfectly describe the study variable. | ✓ 1.00 | |
| ☐ They produce the same answers except for three samples that hierarchical clustering correctly assigns to the right study but k-means does not. | | |
| ☐ They produce different answers, with k-means clustering giving a much more unbalanced clustering. The hierarchical clustering matches study better. | | |
| ☐ They produce different clusterings with hierarchical clustering more closely matching the study variable. K-means clustering is too random to pick up the study difference. | | |
| Total | 1.00 / 1.00 | |